1. Abstract

The Library Management System is a simple database mini-project developed to organize student details, book information, and book issue/return records. The system uses MySQL to store data in three related tables: Student, Book, and IssueBook. Primary keys uniquely identify records, while foreign keys connect issue records with the related student and book. SQL queries are used to display records, search books by name or category, identify currently issued and returned books, count records, and update information. This project demonstrates basic relational database concepts and common SQL operations. A separate HTML/CSS/JavaScript dashboard prototype may be used to demonstrate the user interface; the dashboard prototype uses browser Local Storage unless a backend connection is added.

2. Introduction

A library needs an organized method to maintain information about students, books, and transactions. Keeping these records manually can make searching, updating, and tracking issued books difficult. The Library Management System provides a basic database-based approach to store and retrieve this information.

The project is implemented using MySQL. The Student table stores student ID, name, PRN, department, and phone number. The Book table stores book ID, book name, author, category, and quantity. The IssueBook table stores issue ID, student ID, book ID, issue date, and return date. A NULL return_date indicates that the book has not yet been recorded as returned.

The project is designed as a beginner-friendly DBMS mini-project and demonstrates table creation, insertion, selection, filtering, joins, aggregate functions, updating, and relationships through primary and foreign keys.

3. Course Outcome Integrated

Course Outcome	Description	Project Relevance
CO1	Understand relational database concepts	Three related tables
CO2	Use SQL to define and manipulate data	CREATE, INSERT, UPDATE
CO3	Retrieve data using conditions and joins	SELECT, WHERE, JOIN
CO4	Apply keys and constraints	Primary and foreign keys

4. Action Plan

S. No.	Details of Activity	Planned Start Date	Planned Finish Date	Responsible Team Member
1	Identify requirements	10/08/2026	12/08/2026	Mahesh Ashok Dahihande, Piyush Mukhedkar
2	Design tables and keys	13/08/2026	20/08/2026	Mahesh Ashok Dahihande, Piyush Mukhedkar
3	Write SQL and insert sample data	21/08/2026	15/09/2026	Mahesh Ashok Dahihande, Piyush Mukhedkar
4	Execute queries and test results	16/09/2026	30/09/2026	Mahesh Ashok Dahihande, Piyush Mukhedkar
5	Prepare report and screenshots	01/10/2026	12/10/2026	Mahesh Ashok Dahihande, Piyush Mukhedkar

5. Literature Review

A Library Management System is commonly used to maintain library resources and transaction records. A relational database stores information in tables and uses keys to maintain relationships between records. In this project, Student and Book hold the main entity details, while IssueBook records which student has borrowed which book.

MySQL supports SQL commands for defining database structures and manipulating data. CREATE DATABASE and CREATE TABLE define the database and its tables. INSERT adds records; SELECT retrieves records; WHERE filters results; JOIN combines related records; COUNT calculates the number of rows; UPDATE changes existing values; and DELETE removes records when the related constraints permit it.

Primary keys prevent duplicate identifiers in a table. Foreign keys help maintain referential integrity by ensuring that an issue record refers to an existing student and an existing book. The project uses these concepts to demonstrate a basic relational database design.

6. Proposed Methodology

The project follows a simple database development process. First, the library database is created in MySQL. Next, the Student, Book, and IssueBook tables are created with suitable data types and key constraints. Sample student, book, and issue records are inserted into the tables.

SQL SELECT queries display all students, books, and issue records. Search queries retrieve a particular book or books belonging to a selected category. JOIN queries combine student, issue, and book information. A condition on return_date separates currently issued books from returned books. COUNT queries report the number of student records and book titles. UPDATE statements demonstrate changes to book quantity and student phone number. Additional INSERT statements demonstrate adding a new book and a new student.

Workflow: Start → Create/Use database → Create tables → Insert sample records → Run search and report queries → Update or add records → Review results.

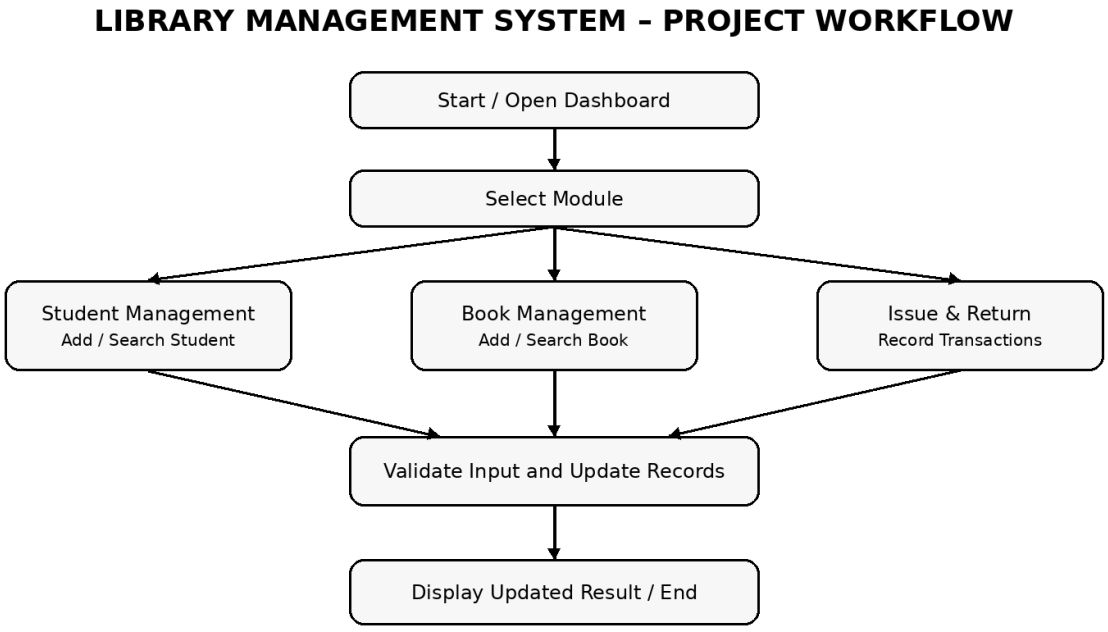


Figure 1: Workflow of the Library Management System

7. Implementation

The database is named library. It contains the following tables:

Student(student_id, name, prn, department, phone)

Book(book_id, book_name, author, category, quantity)

IssueBook(issue_id, student_id, book_id, issue_date, return_date)

The student_id column in IssueBook references Student(student_id), and book_id references Book(book_id). The SQL script inserts five students, six book titles, and three issue records. It then runs display, search, join, filtering, counting, updating, and insertion queries.

Important implementation note: the supplied SQL script changes the quantity of Java Programming to 6 as an example update, but it does not automatically decrement or increment quantity when an issue or return is recorded. Automatic stock handling would require additional SQL statements, triggers, stored procedures, or application logic.

8. Output of Mini Project

The expected output consists of result sets showing the records stored in the three tables and the results of the search, join, count, update, and insert queries. Based on the inserted sample data, the database initially contains 5 student records, 6 book-title records, and 3 issue records. Two issue records have NULL return dates and are therefore shown as currently issued; one record has a return date and is shown as returned.

After the UPDATE statements run, the quantity for book_id 101 is set to 6 and the phone number for student_id 1 is set to 9999999999. The later INSERT statements add book_id 107 (Web Development) and student_id 6 (Akash). Because the script is sequential, the final database after all statements contains 6 students and 7 book-title records.

Insert actual screenshots from MySQL Workbench or the MySQL command line here if your college requires screenshots of executed queries and result grids.

Query / Operation	Expected Result	Purpose
SELECT * FROM Student	5 initial rows	Display students
SELECT * FROM Book	6 initial rows	Display books
Currently issued query	2 rows	Find books not yet returned
Returned books query	1 row	Find returned books
COUNT(*) on Student	5 initially; 6 after final INSERT	Count students
COUNT(*) on Book	6 initially; 7 after final INSERT	Count book titles

9. Skill Developed / learning out of this Micro-Project

Through this mini-project, we learned how to create a database and tables in MySQL, select appropriate data types, define primary and foreign keys, and insert sample records. We practiced SELECT queries, WHERE conditions, JOIN operations, COUNT functions, and UPDATE statements. We also learned how relational tables can be connected to represent the relationship between students, books, and issue transactions.

The project helped us understand how to organize data, retrieve information through queries, and test database operations using sample records.

10. Applications of this Micro-Project

This project can be used as a basic model for a college library, school library, department book collection, or small reading room. Its table design can be extended to track members, books, and borrowing transactions.

The same database concepts can also be adapted for inventory systems, equipment lending, laboratory asset records, and other applications in which items are assigned to users and their transactions must be stored.

11. Area of Future Improvement

The project can be improved by adding a graphical user interface connected to MySQL through a backend such as Java with JDBC. User login and role-based access can be added for librarians and students. The system can validate duplicate PRNs, invalid phone numbers, unavailable books, and incorrect dates.

Further improvements may include automatic quantity updates during issue and return operations, due-date tracking, overdue fine calculation, book reservation, reports by date or category, and backup/restore features. The dashboard prototype should connect to the actual MySQL database through a backend before it is described as a live database-connected application. Screenshots of the real program output can be inserted into the final report.

12. Conclusion

The Library Management System mini-project demonstrates the use of MySQL to maintain student information, book details, and issue/return records. It applies important DBMS concepts including relational tables, primary keys, foreign keys, insertion, retrieval, filtering, joins, aggregate functions, and updating data. The project is simple enough for beginners and can be expanded into a complete application by connecting a user interface to the database and adding automatic issue/return processing.